1. Propose

The Restaurant Table Reservation System (RTS) is a web-based application that allows customers to reserve restaurant tables online while enabling restaurant staff and administrators to manage reservations, table availability, and restaurant information efficiently.

This specification is intended for software development and software testing activities.

2. Scope

The system shall allow:

- Customers to create and manage reservations online.
- Staff to manage daily reservations and table status.
- Administrators to configure restaurant information and monitor system usage.
- Automatic updating of table availability in real time.

3. Definitions

Term	Description
Customer	User who makes reservations
Staff	Employee managing daily operations
Admin	System administrator with configuration privileges
Reservation	Booking of a table for a specific date and time
Table	Physical seating unit with fixed capacity

4. User Roles (Resolved Inconsistency)

4.1 Customer

- Register and log in
- Make, modify, cancel reservations
- View reservation history

4.2 Staff

- View reservations

- Create walk-in reservations
- Update table status

4.3 Admin

- Manage restaurant profile
- Manage tables
- View reports and statistics

(*The undefined "Restaurant Management" role is merged into Staff/Admin.*)

5. Functional Requirements

5.1 Account Management

FR-01 Registration

- The system shall allow customers to register using:
  - Name
  - Email (unique)
  - Phone number
  - Password

Field	Type	Required	Length	Validation Rule
email	string	Yes	5–100 chars	Must be valid email format
password	string	Yes	8–20 chars	At least 1 uppercase, 1 number
fullName	string	Yes	2–60 chars	Letters and spaces only
phoneNumber	string	No	10–15 chars	Digits only

FR-02 Login

- The system shall authenticate users using email and password.
- The system shall display an error message for invalid credentials.

5.2 Reservation Management

FR-03 View Availability

- The system shall display available tables based on:
  - Date
  - Time

- Number of guests.

FR-04 Make Reservation

The system shall allow customers to create a reservation including:

- Date
- Time
- Number of guests
- Optional special request.

Field	Type	Required	Range / Length	Example
date	date (YYYY-MM-DD)	Yes	future date only	2026-04-10
time	time (HH:mm)	Yes	restaurant hours only	18:00
tableSize	integer	Yes	1–10 persons	4
specialRequest	string	No	0–200 chars	window seat

Validation rules:

- Reservation must be within operating hours.
- Guest count must not exceed table capacity.
- Past dates shall not be allowed.

FR-05 Reservation Confirmation

- The system shall display confirmation immediately after successful booking.
- The system shall store reservation data in the Reservation Database.

FR-06 Modify Reservation

- Customers shall modify reservations before the reservation time.

FR-07 Cancel Reservation

- Customers may cancel reservations at least 1 hour before reservation time.
- The system shall automatically update table availability after cancellation.

FR-08 Notification

- The system shall notify staff via dashboard notification when:

    - a reservation is created
    - modified
    - cancelled.

5.3 Table Management

FR-09 Table Status

Staff shall update table status as:

- Available
- Occupied
- Out of Service.

FR-10 Double Booking Prevention

- The system shall prevent multiple reservations for the same table and time slot.

5.4 Walk-in Reservation

FR-11 Manual Reservation

- Staff shall create reservations for walk-in customers.

5.5 Restaurant Profile Management

FR-12 Manage Restaurant Information

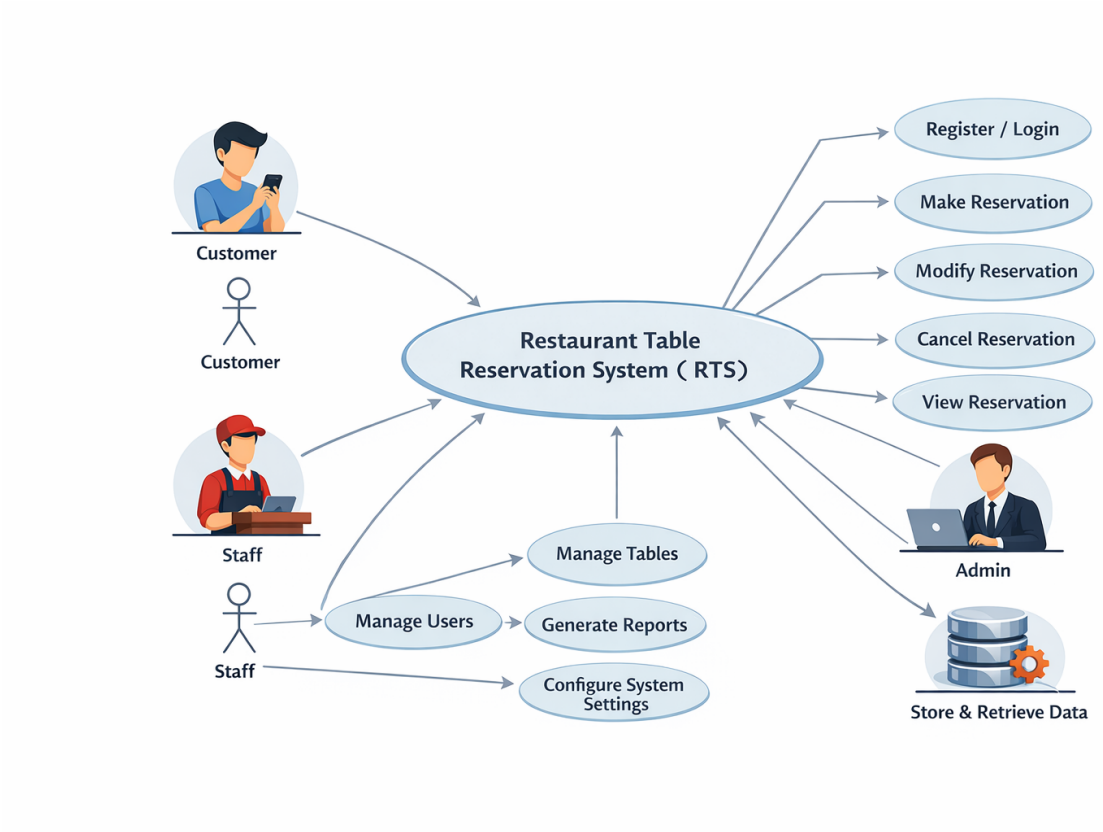
Admin shall update:

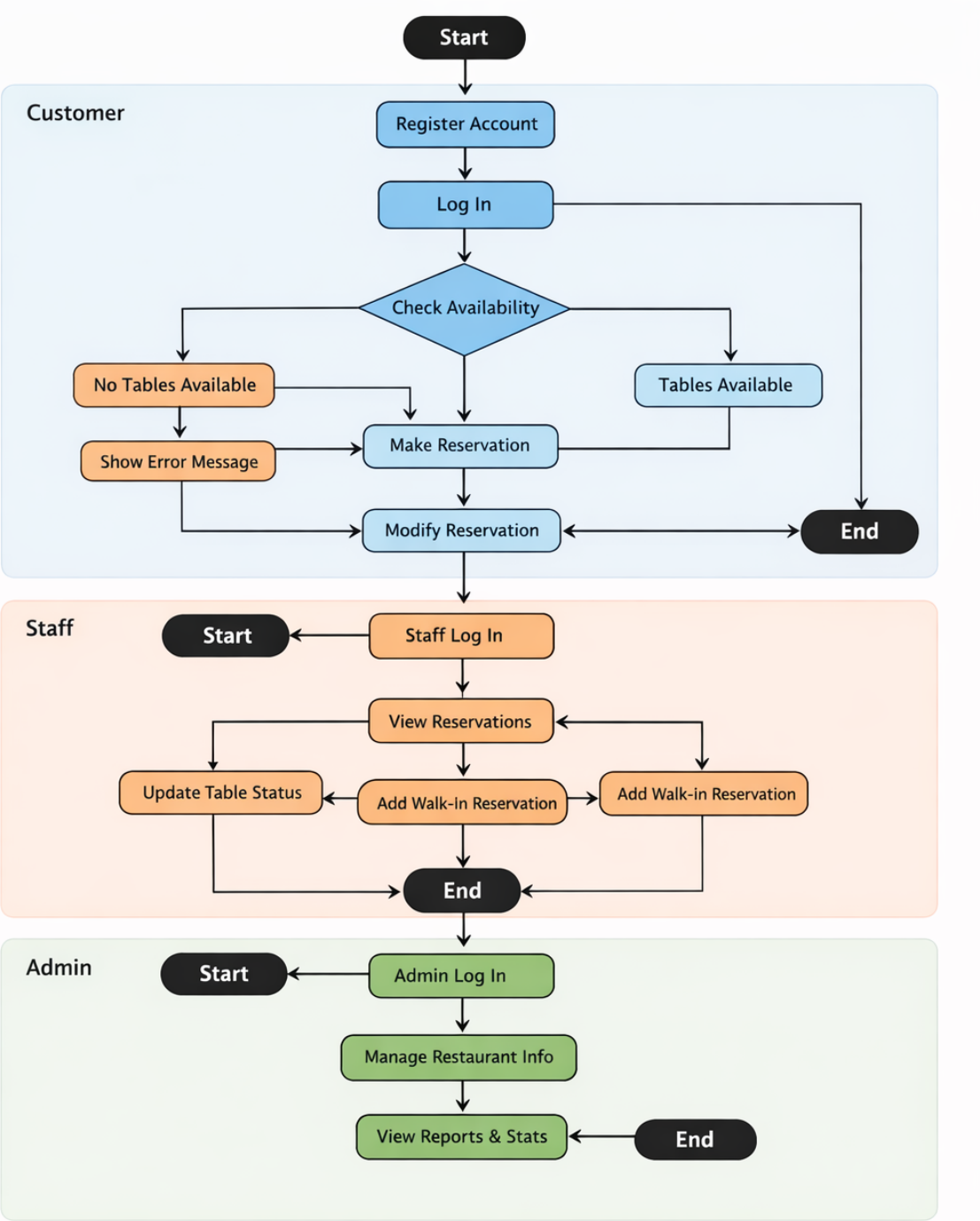
- Operating hours
- Restaurant name
- Contact information.

Information shall be stored in the Restaurant Database.

6. Data Flow

Use case design




Start
Customer
Register Account
Log In
Check Availability
No Tables Available
Tables Available
Show Error Message
Make Reservation
Modify Reservation
End
Staff
Start
Staff Log In
View Reservations
Update Table Status
Add Walk-in Reservation
Add Walk-in Reservation
End
Admin
Start
Admin Log In
Manage Restaurant Info
View Reports & Stats
End

7. Performance Testing

7.1 Response Time Requirement

API	Max Response Time
POST /auth/login	≤ 1 second
POST /reservations	≤ 2 seconds
GET /reservations/my	≤ 1.5 seconds
DELETE /reservations/{id}	≤ 1 second
GET /tables	≤ 1 second

7.2 Concurrent Users

System must support: 1000 concurrent users

7.3 Error Rate

Error rate must be < 1%

7.4 Duration

System should run 1 hour smoothly

8. API Specification (Student-Friendly Version)

8.1 Base Information

Base URL: https://api.restaurant-reservation.com/v1

Data Format: Request and Response use JSON

Authentication: Bearer Token (JWT)

8.2 User Roles

Role	Description
Customer	Make and manage reservations
Staff	Manage tables and reservations
Admin	Full system control

8.3 Authentication APIs

8.3.1 Register Customer

Method: POST

Endpoint: /auth/register

Request Example:

```
{
  "name": "John Smith",
  "email": "john@email.com",
  "password": "Password123",
  "phone": "0812345678"
}
```

Response Example:

```
{
  "userId": 101,
  "message": "Registration successful"
}
```

Method: POST

Endpoint: /auth/login

Request Example:

```
{
  "email": "john@email.com",
  "password": "Password123"
}
```

Response Example:

```
{
  "token": "jwt_token_here",
  "role": "Customer"
}
```

8.4 Reservation APIs

8.4.1 Create Reservation

Method: POST

Endpoint: /reservations

Request Example:

```
{
  "date": "2026-05-20",
  "time": "18:00",
  "guestCount": 4,
  "specialRequest": "Window seat"
}
```

Response Example:

```
{
  "reservationId": 5001,
  "status": "CONFIRMED"
}
```

Business Rules:

- Guest count must be between 1 and 10
- Reservation must be future date/time
- Table must be available

8.4.2 Cancel Reservation

Method: DELETE

Endpoint: /reservations/{reservationId}

Response Example:

```
{
  "message": "Reservation cancelled"
}
```

8.5 Table Management (Staff)

8.5.1 Get Table List

Method: GET

Endpoint: /tables

Response Example:

```
{
  "tableId": 1,
  "capacity": 4,
  "status": "AVAILABLE"
}
```

8.6 Admin APIs

8.6.1 View All Reservations

Method: GET

Endpoint: /admin/reservations

Response Example:

```
{
  "reservationId": 5001,
  "customerName": "John Smith",
  "tableId": 3,
  "status": "CONFIRMED"
}
```

8.7 Error Responses

HTTP Code	Meaning
400	Invalid input data
401	Authentication required
404	Reservation not found
409	Table not available

9. Security Test

9.1 Password Policy

9.2 Login Attempt Limitation

9.3 Token Expiration

9.4 Input Validation Security

9.5 API Security

9.6 Sensitive data protection

Appendix

This specification will be used by students to perform software testing activities such as:

- Requirement analysis
- Test case design
- Functional testing
- Boundary testing
- Performance testing
- Security testing

No.	Role	Responsibility
1	QA Lead / BA	Requirement Analysis + Report
2	Test Designer	Design Test Cases
3	Dev in Test	Unit Test
4	API Tester	API Test
5	Performance Tester	Performance Test
6	Security Tester	Security Test